6 Lina and Mona are two statisticians who also write songs. The 'time' of a song is the number of minutes for which it lasts. For a random sample of 10 of her songs, Lina calculates a 95% confidence interval for the population mean time, μ minutes. This confidence interval is $2.95 \leqslant \mu \leqslant 3.13$. The times, x minutes, of Lina's songs are normally distributed.

(a) Find the values of $\sum x$ and $\sum x^2$ for the 10 songs in Lina's sample. [5]

Mona's songs have times, y minutes, that are normally distributed. The times for a random sample of 8 of Mona's songs are summarised as follows.

$$\sum y = 24.8 \qquad \sum y^2 = 76.98$$

Mona claims that the population mean time of her songs is greater than the population mean time of Lina's songs.

(b) Assuming that the two distributions have the same population variance, test at the 5% significance level whether there is evidence to support Mona's claim. [8]